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Immersion cooling system

Abstract

An immersion cooling system includes a work tank, a chip device, a microchannel device, a first communication pipeline, and a first heat exchange device. The work tank includes a fluid section. The chip device is in the fluid section and has an inlet and an outlet. The chip device includes a motherboard, a chip, and a cover. The motherboard has a main surface substantially parallel to a vertical line. The cover is on the motherboard. The chip is between the cover and the motherboard. The microchannel device is in the chip device. Two ends of the microchannel device are respectively in communication with the inlet and the outlet. The first communication pipeline and the first heat exchange device are in the fluid section. Two ends of the first communication pipeline are respectively in communication with the outlet and the first heat exchange device.

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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. provisional application Ser. No. 63/398,985, filed on Aug. 18, 2022 and claims the priority of Patent Application No. 112113736 filed in Taiwan, R.O.C. on Apr. 12, 2023. The entirety of the above-mentioned patent applications are hereby incorporated by references herein and made a part of the specification.

BACKGROUND

Technical Field

(1) The instant disclosure relates to an immersion cooling system, particularly to an immersion cooling system having a microchannel device.

Related Art

(2) In the immersion cooling system known to the inventor, component(s) to be cooled is usually placed in the working fluid, and a cooling device outside the system (or outside the work tank of the system) is in communication with the component(s) to be cooled to cool the component(s) to be cooled.

SUMMARY

(3) Since the cooling device is outside the system, the cooling device is in communication with the component(s) inside the system (or inside the work tank of the system) from the exterior of the system (or the exterior of the work tank of the system). Therefore, a relative longer communication pipeline is needed for such cooling device, and thus a higher pumping power is needed to be consumed to achieve a desired cooling performance of the system.

(4) Accordingly, in some embodiments, an immersion cooling system comprising a work tank, a chip device, a microchannel device, a first communication pipeline, and a first heat exchange device is provided. The work tank comprises a fluid section. The chip device is in the fluid section, and the chip device has an inlet and an outlet. The chip device comprises a motherboard, a chip, and a cover. The motherboard has a main surface substantially parallel to a vertical line. The cover is on the motherboard. The chip is between the cover and the motherboard. The microchannel device is in the chip device. A first end of the microchannel device is in communication with the inlet, and a second end of the microchannel device is in communication with the outlet. The first communication pipeline is in the fluid section, and a first end of the first communication pipeline is in communication with the outlet. The first heat exchange device is in the fluid section and in communication with a second end of the first communication pipeline.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates a schematic perspective view of an immersion cooling system according to a first embodiment;

(2) FIG. 2 illustrates a schematic perspective view of a partial section A of the immersion cooling system shown in FIG. 1 according to a second embodiment;

(3) FIG. 3 illustrates a schematic perspective view of a partial section A of the immersion cooling system shown in FIG. 1 according to a third embodiment;

(4) FIG. 4 illustrates a schematic perspective view of a partial section A of the immersion cooling system shown in FIG. 1 according to a fourth embodiment;

(5) FIG. 5 illustrates a schematic perspective view of a partial section A of the immersion cooling system shown in FIG. 1 according to a fifth embodiment;

(6) FIG. 6 illustrates a schematic perspective view of a partial section A of the immersion cooling system shown in FIG. 1 according to a sixth embodiment;

(7) FIG. 7 illustrates a schematic perspective view of a partial section A of the immersion cooling system shown in FIG. 1 according to a seventh embodiment;

(8) FIG. 8A illustrates a cross-sectional schematic view of the first communication pipeline according to some embodiments;

(9) FIG. 8B illustrates a cross-sectional schematic view of the first communication pipeline according to some embodiments;

(10) FIG. 8C illustrates a cross-sectional schematic view of the first communication pipeline

according to some embodiments;

(11) FIG. 9A illustrates a cross-sectional schematic view of the cooling device according to some embodiments;

(12) FIG. 9B illustrates a cross-sectional schematic view of the cooling device according to some embodiments; and

(13) FIG. 9C illustrates a cross-sectional schematic view of the cooling device according to some embodiments.

DETAILED DESCRIPTION

(14) Please refer to FIG. 1. FIG. 1 illustrates a schematic perspective view of an immersion cooling system 1 according to a first embodiment. In FIG. 1, an immersion cooling system 1 comprises a work tank 10, a chip device 2, a microchannel device 11, a first communication pipeline 12, and a first heat exchange device 13. The work tank 10 is adapted to receive a working fluid (e.g., the liquid-phase working fluid 101, which will be described later) and comprises a fluid section L and a vapor section V. The fluid section L is adapted to receive a liquid-phase working fluid 101, and the vapor section V is adapted to receive a mixed gas-phase fluid 102 (which comprises a gas-phase working fluid and/or moisture mixed therein, and will be described later). The chip device 2 is in the fluid section L and has an inlet 200 and an outlet 201 (which will be described later). The chip device 2 comprises a motherboard 20, a chip 21, and a cover 22. The motherboard 20 has a main surface 202, and the main surface 202 is substantially parallel to a vertical line (e.g., the Z direction shown in FIG. 1); that is, in the case that the main surface 202 is substantially parallel to the vertical line, an acute angle may be between the main surface 202 and the vertical line (e.g., the Z direction shown in FIG. 1), and the acute angle is, for example, less than 45 degrees. The number of the chip 21 may be one or more, which is not limited herein. For example, a plurality of the chips 21 may be three-dimensional integrated (e.g., integrated along a direction perpendicular to the main surface 202; for example, the stacking direction D1 shown in FIG. 1) to form a chip-stacked structure (or referred to as a chip-integrated structure). The cover 22 is on the motherboard 20 (which will be described later). The chip(s) 21 is between the cover 22 and the motherboard 20; for example, a closed space is formed between the cover 22 and the motherboard 20 to further receive the chip(s) 21. The microchannel device 11 is in the chip device 2. A first end of the microchannel device 11 is in communication with the inlet 200, and a second end of the microchannel device 11 is in communication with the outlet 201 (which will be described later). The first communication pipeline 12 is in the fluid section L, and a first end of the first communication pipeline 12 is in communication with the outlet 201 (which will be described later). The materials of the first communication pipeline 12 may comprise at least one selected from the group consisting of metal, alloy, ceramic, plastic, and rubber, which is not limited herein. The first heat exchange device 13 is in the fluid section L, and the first heat exchange device 13 is in communication with a second end of the first communication pipeline 12 (which will be described later). Hence, by configuring the chip device 2 to be in communication with the liquid-phase working fluid 101 and the microchannel device 11, the chip device 2 can be cooled sufficiently. Furthermore, through the first communication pipeline 12 and the first heat exchange device 13 that are both in the fluid section L and in communication with the chip device 2, problems encountered by the cooling device known to the inventor (which is outside the system) can be avoided. For example, the following problems can be avoided, such as the needs of longer communication pipelines and even a higher pumping power to achieve the desired cooling performance.

(15) Please still refer to FIG. 1. In some embodiments, the work tank 10 is adapted to receive a working fluid and the component(s) to be cooled. The component(s) to be cooled may be any device having a temperature higher than the temperature of the working fluid (e.g., the liquid-phase working fluid 101); for example, the component(s) to be cooled may be the chip device 2 shown in FIG. 1. The working fluid (or referred to as the heat transfer fluid) is a non-conductive fluid. Under

normal circumstances (i.e., at a temperature lower than the boiling point of the working fluid), the working fluid is a liquid (i.e., the liquid-phase working fluid **101**), and the boiling point of the working fluid is lower than or substantially equal to the temperature of the component(s) to be cooled. Hence, when the component(s) to be cooled is immersed in the liquid-phase working fluid **101**, the liquid-phase working fluid **101** may absorb the heat from the component(s) to be cooled and thus the temperature of the liquid-phase working fluid **101** reaches its critical boiling point rapidly, so that the liquid-phase working fluid **101** is further evaporated as a gas-phase working fluid.

(16) Please still refer to FIG. **1**. In some embodiments, the first heat exchange device **13** may be a boiler plate. Specifically, in some embodiments, the first heat exchange device **13** comprises a partition plate **130** and a plurality of heat exchange channels **131**. The partition plate **130** is adapted to partition the heat exchange channels **131**. The heat exchange channels **131** are correspondingly in communication with the second end of the first communication pipeline **12**. The microchannel device **11** and the first heat exchange device **13** (and the heat exchange channels **131** of the first heat exchange device **13**) are adapted to be applied with a circulation fluid **120**, and the circulation fluid **120** may be a fluid identical to or different from the liquid-phase working fluid **101**, which is not limited herein. In some embodiments, the diameter of each of the heat exchange channels **131** is less than the diameter of the second end of the first communication pipeline **12**. Hence, the circulation fluid **120** inside the first communication pipeline **12** may be distributed over the heat exchange channels **131**, so that the distributed circulation fluid **120** may conduct heat exchange synchronously and more sufficiently. The materials of the partition plate **130** and the heat exchange channels **131** may independently be any material with thermal conductive function, which is not limited herein; for example, the materials may be metal, alloy, or a combination thereof.

(17) In some embodiments, the immersion cooling system **1** further comprises a second heat exchange device **18**, wherein the second heat exchange device **18** is in the vapor section V. For example, still referring to FIG. **1**, the second heat exchange device **18** comprises a condenser **180**, wherein the condenser **180** is in the vapor section V; for example, the condenser **180** is located above the surface of the liquid-phase working fluid **101**. The condenser **180** may be applied with a cooling fluid, and both the condenser **180** and the cooling fluid of the condenser **180** have a temperature lower than the temperature of the mixed gas-phase fluid **102**; for example, a temperature lower than the dew point or the boiling point of the working fluid. Therefore, when the gas-phase working fluid in the mixed gas-phase fluid **102** contacts the condenser **180**, a heat exchange between the gas-phase working fluid and the condenser **180** may be conducted. The temperature of the gas-phase working fluid would be then reduced, and the gas-phase working fluid would be condensed into the liquid-phase working fluid **101** to flow back to the fluid section L and be reused for cooling the component(s) to be cooled (e.g., the chip device **2** shown in FIG. **1**). Accordingly, through the liquid-phase working fluid **101** reused in the closed environment, the required usage of the liquid-phase working fluid **101** can be prevented from being unnecessarily increased, thereby enhancing the cooling performance provided by per unit of the liquid-phase working fluid **101**.

(18) Please still refer to FIG. **1**. In some embodiments, the second heat exchange device **18** further comprises a condensation pump **182** and a heat exchanger **184**. The condensation pump **182** and the heat exchanger **184** are independently inside or outside the work tank **10**, which is not limited herein. For example, in FIG. **1**, the condensation pump **182** and the heat exchanger **184** are both outside the work tank **10**, and only the condenser **180** is inside the work tank **10** (e.g., in the vapor section V of the work tank **10**). The condensation pump **182** may be any device that can increase the pressure of the fluid inside the device, which is not limited herein. A first end of the condensation pump **182** is in communication with a first end of the condenser **180** (for example, through a first condensation pipe **181**). A second end of the condensation pump **182** is in communication with a first end of the heat exchanger **184** (for example, through a second

condensation pipe **183**). A second end of the heat exchanger **184** is in communication with a second end of the condenser **180** (for example, through a third condensation pipe **185**). The materials of the first condensation pipe **181**, the second condensation pipe **183**, and the third condensation pipe **185** may independently comprise at least one selected from the group consisting of metal, alloy, ceramic, plastic, and rubber, which is not limited herein. Accordingly, through a condensation circulation formed by the condenser **180**, the condensation pump **182**, and the heat exchanger **184**, the cooling fluid from the condenser **180** (with a higher temperature) can be cooled by decreasing the temperature of the cooling fluid to be lower than the dew point or the boiling point of the working fluid, so that the cooling fluid is pumped back to the condenser **180** to condense the gas-phase working fluid. Hence, the liquid-phase working fluid **101** in the work tank **10** can be recycled and reused for cooling the component(s) to be cooled (e.g., the chip device **2** shown in FIG. **1**).

(19) In some embodiments, the inlet **200** is selectively on at least one of the motherboard **20** and the cover **22**; and the outlet **201** is selectively on at least one of the motherboard **20** and the cover **22**. In other words, in some embodiments, the inlet **200** is selectively on the motherboard **20**, or on the cover **22**, or on both the motherboard **20** and the cover **22**; and the outlet **201** is selectively on the motherboard **20**, or on the cover **22**, or on both the motherboard **20** and the cover **22**.

(20) For example, referring to both FIG. **1** and FIG. **4**, FIG. **4** illustrates a schematic perspective view of a partial section A of the immersion cooling system **1** shown in FIG. **1** according to a fourth embodiment. In FIG. **4**, the inlet **200** is on the motherboard **20**, and the outlet **201** is also on the motherboard **20**. The microchannel device **11** comprises a first microchannel **110** and a second microchannel **111**. The circulation fluid **120** can enter the second microchannel **111** from the inlet **200** (along an entering direction DO) and be discharged from the first microchannel **110**, so that the chip device **2** (shown in FIG. **1**) and the chip(s) **21** of the chip device **2** can be cooled sufficiently by the circulation fluid **120**.

(21) Specifically, still referring to FIG. **1** and FIG. **4**, in some embodiments, the number of the first microchannel **110** may be one or more, and the first communication pipeline **12** is in communication with the first microchannel(s) **110** through the outlet(s) **201**. For example, a first end of the first microchannel **110** is in communication with the chip(s) **21**, a second end of the first microchannel **110** is in communication with the outlet **201**, and the diameter of the first microchannel **110** is less than the diameter of the first communication pipeline **12**. Furthermore, the first end of each of the first microchannels **110** may be configured between one or more chips **21** according to the patterns and/or the stacking of the chips **21**. The number of the second microchannel **111** may be one or more, and each of the second microchannels **111** is correspondingly in communication with the inlet **200**. For example, a first end of the second microchannel **111** is in communication with the chip(s) **21**, and a second end of the second microchannel **111** is in communication with the inlet **200**. Furthermore, the first end of each of the second microchannels **111** may be configured between one or more chips **21** according to the patterns and/or the stacking of the chips **21**.

(22) The first microchannel **110** and the second microchannel **111** in the chip device **2** (e.g., at least in the chip(s) **21**) are substantially in communication with each other, and the circulation fluid **120** can flow inside the first microchannel **110** and the second microchannel **111**. The materials of the first microchannel **110** and the second microchannel **111** may independently be plastic, rubber, and a combination thereof, which is not limited herein. Hence, through the circulation fluid **120** inside the first microchannel **110** and the second microchannel **111**, the chip(s) **21** can be cooled sufficiently. In this embodiment, the circulation fluid **120** may be a fluid identical to the liquid-phase working fluid **101** (shown in FIG. **1**), and thus the contamination between the fluids in the work tank **10** can be avoided. Accordingly, in some embodiments, the inlet **200** is not necessarily needed to be in communication with other communication pipelines (e.g., the second communication pipeline **14**) and/or the pump **15** (which will be described later). Due to the

capillary action and/or thermal convection, the circulation fluid **120** (that is, the liquid-phase working fluid **101** in this embodiment) can still enter the second microchannel **111** through the inlet **200**, so that the chip device **2** (shown in FIG. **1**) and the chip(s) **21** of the chip device **2** can be cooled sufficiently.

(23) For another example, referring to both FIG. **1** and FIG. **3**, FIG. **3** illustrates a schematic perspective view of a partial section A of the immersion cooling system **1** shown in FIG. **1** according to a third embodiment. In FIG. **3**, the inlet **200** is on the motherboard **20**, and the outlet **201** is also on the motherboard **20**. Moreover, the immersion cooling system **1** further comprises a second communication pipeline **14**, wherein the second communication pipeline **14** is in the fluid section L, the second communication pipeline **14** is in communication with the inlet **200**, and the diameter of the second microchannel **111** is less than the diameter of the second communication pipeline **14**. The materials of the second communication pipeline **14** may comprise at least one selected from the group consisting of metal, alloy, ceramic, plastic, and rubber, which may be identical to or different from the materials of the first communication pipeline **12** and is not limited herein. In this embodiment, the circulation fluid **120** may be a fluid identical to the liquid-phase working fluid **101** (shown in FIG. **1**), and thus the contamination between the fluids in the work tank **10** can be avoided. Accordingly, in some embodiments, the inlet **200** and the second communication pipeline **14** are not necessarily needed to be in communication with other pumps **15** (which will be described later). Due to the capillary action and/or thermal convection, the circulation fluid **120** (that is, the liquid-phase working fluid **101** in this embodiment) can still enter the second microchannel **111** through the second communication pipeline **14** and the inlet **200**, so that the chip device **2** (shown in FIG. **1**) and the chip(s) **21** of the chip device **2** can be cooled sufficiently. Furthermore, through the second communication pipeline **14** that is also in the fluid section L of the work tank **10** (shown in FIG. **1**) and in communication with the chip device **2**, problems encountered by the cooling device known to the inventor (which is outside the system) can be avoided by the immersion cooling system **1** according to some embodiments.

(24) For another example, referring to both FIG. **1** and FIG. **2**, FIG. **2** illustrates a schematic perspective view of a partial section A of the immersion cooling system **1** shown in FIG. **1** according to a second embodiment. In FIG. **2**, the inlet **200** is on the motherboard **20**, and the outlet **201** is also on the motherboard **20**. Moreover, the immersion cooling system **1** further comprises a pump **15**, wherein the pump **15** is in the fluid section L, and the pump **15** is in communication with the second communication pipeline **14**. For example, in FIG. **2**, two ends of the second communication pipeline **14** are respectively in communication with the pump **15** and the inlet **200**. In this embodiment, the circulation fluid **120** may be a fluid identical to the liquid-phase working fluid **101** (shown in FIG. **1**), and thus the contamination between the fluids in the work tank **10** can be avoided. The condensation pump **182** may be any device that can increase the pressure of the fluid inside the device, which is not limited herein. In addition to the capillary action and/or thermal convection, due to the pumping of the pump **15**, the circulation fluid **120** (that is, the liquid-phase working fluid **101** in this embodiment) can enter the second microchannel **111** through the second communication pipeline **14** and the inlet **200**, so that the chip device **2** (shown in FIG. **1**) and the chip(s) **21** of the chip device **2** can be cooled more rapidly and sufficiently. Furthermore, through the pump **15** that is also in the fluid section L of the work tank **10** (shown in FIG. **1**) and in communication with the chip device **2** and the second communication pipeline **14**, problems encountered by the cooling device known to the inventor (which is outside the system) can be avoided by the immersion cooling system **1** according to some embodiments.

(25) In some embodiments, the chip device **2** is electrically connected to the pump **15** to supply electricity for the pump **15**, and the chip device **2** may transmit a control signal to pump **15** so as to control the pump **15** (for example, to actuate the pump **15** for pumping the circulation fluid **120**). In some embodiments, the pump **15** is on the motherboard **20** of the chip device **2** so as to be electrically connected to the motherboard **20** and to receive the control signal from the chip device

2 and the chip(s) 21 of the chip device 2. Accordingly, the pump 15 is not necessarily needed to be connected to an additional power supply and/or control device (e.g., an IPC), and thus an unnecessary increase on the cost due to the contamination of the liquid-phase working fluid 101 can be avoided.

(26) For another example, still referring to FIG. 1, in FIG. 1, the inlet 200 is on the motherboard 20, and the outlet 201 is also on the motherboard 20. Moreover, the immersion cooling system 1 further comprises a third communication pipeline 16, wherein the third communication pipeline 16 is in the fluid section L, a first end of the third communication pipeline 16 is in communication with the first heat exchange device 13, and a second end of the third communication pipeline 16 is in communication with the pump 15. In this embodiment, due to a closed circulation formed by the chip device 2, the microchannel device 11, the first communication pipeline 12, the first heat exchange device 13, the third communication pipeline 16, the pump 15, and the second communication pipeline 14, the circulation fluid 120 may be a fluid identical to or different from the liquid-phase working fluid 101, which is not limited herein. Accordingly, the circulation fluid 120 can be independently discharged from the outlet 201 so as to conduct heat exchange with the first heat exchange device 13 and pumped by the pump 15, and the circulation fluid 120 is further transported back to the second microchannel 111 through the second communication pipeline 14 and the inlet 200, so that the chip device 2 and the chip(s) 21 of the chip device 2 can be cooled sufficiently. Furthermore, in addition to the pump 15 and the second communication pipeline 14, through the third communication pipeline 16 that is also in the fluid section L of the work tank 10 and in communication with the pump 15 and the first heat exchange device 13, problems encountered by the cooling device known to the inventor (which is outside the system) can be avoided by the immersion cooling system 1 according to some embodiments.

(27) For another example, referring to both FIG. 1 and FIG. 6, FIG. 6 illustrates a schematic perspective view of a partial section A of the immersion cooling system 1 shown in FIG. 1 according to a sixth embodiment. In FIG. 6, the inlet 200 is on the cover 22, and the outlet 201 is also on the cover 22. The materials of the cover 22 may be any material with thermal conductive function, which is not limited herein. Moreover, in this embodiment, the immersion cooling system 1 may optionally comprise the second communication pipeline 14, the pump 15, and the third communication pipeline 16; detail configurations of the implementations can be referred to above descriptions and thus will not be further described in detail herein. Therefore, through various configurations of the inlet 200 and the outlet 201, according to some embodiments, the immersion cooling system 1 that can prevent from the problems encountered by the cooling device known to the inventor (which is outside the system) is provided. Furthermore, through the cover 22 that is also in the fluid section L and has thermal conductive function, the heat of the chip device 2 (shown in FIG. 1) can be thermally conducted outward by the cover 22, and the heat then conducts heat exchange with the liquid-phase working fluid 101, so that the chip device 2 (shown in FIG. 1) and the chip(s) 21 of the chip device 2 can be cooled.

(28) For another example, referring to both FIG. 1 and FIG. 7, FIG. 7 illustrates a schematic perspective view of a partial section A of the immersion cooling system 1 shown in FIG. 1 according to a seventh embodiment. In FIG. 7, the inlet (including the inlet 200 and the inlet 200') is on both the motherboard 20 and the cover 22 at the same time (that is, in this embodiment, the inlet 200 is on the motherboard 20, and the inlet 200' is on the cover 22), and the outlet (including the outlet 201 and the outlet 201') is on both the motherboard 20 and the cover 22 at the same time (that is, in this embodiment, the outlet 201 is on the motherboard 20, and the outlet 201' is on the cover 22). The microchannel device 11 comprises a plurality of first microchannels 110, 110' and a plurality of second microchannels 111, 111'. The circulation fluid 120 may enter the second microchannel 111 from the inlet 200 (along the entering direction D0) and be discharged from the first microchannel 110, while the circulation fluid 120 may also enter the second microchannel 111' from the inlet 200' (along the entering direction D0') and be discharged from the first microchannel

110'. Therefore, the chip device **2** (shown in FIG. **1**) and the chip(s) **21** of the chip device **2** can be cooled efficiently.

(29) Specifically, still referring to FIG. **1** and FIG. **7**, the first communication pipeline **12** is in communication with the first microchannel **110** through the outlet **201**, and the first communication pipeline **12** is in communication with the first microchannel **110'** through the outlet **201'**. For example, the first end of each of the first microchannels **110**, **110'** is in communication with the chip(s) **21**, the second ends of the first microchannels **110**, **110'** are respectively in communication with the outlets **201**, **201'**, and the diameter of each of the first microchannels **110**, **110'** is less than the diameter of the first communication pipeline **12**. Furthermore, the first end of each of the first microchannels **110**, **110'** may be configured between one or more chips **21** according to the patterns and/or the stacking of the chips **21**. The second microchannel **111** is in communication with the inlet **200**, and the second microchannel **111'** is in communication with the inlet **200'**. For example, the first end of each of the second microchannels **111**, **111'** is in communication with the chip(s) **21**, and the second ends of the second microchannels **111**, **111'** are respectively in communication with the inlets **200**, **200'**. Furthermore, the first end of each of the second microchannels **111**, **111'** may be configured between one or more chips **21** according to the patterns and/or the stacking of the chips **21**. The materials of the first microchannel **110** and the first microchannel **110'** may be identical to or different from each other, and the materials of the second microchannel **111** and the second microchannel **111'** may be identical to or different from each other, both of which are not limited herein. Moreover, in this embodiment, the immersion cooling system **1** may comprise the second communication pipeline **14**, or comprise the second communication pipeline **14** and the pump **15**, or comprise the second communication pipeline **14**, the pump **15**, and the third communication pipeline **16**; detail configurations of the implementations can be referred to above descriptions and thus will not be further described in detail herein. Therefore, through various configurations of the inlets **200**, **200'** and the outlets **201**, **201'**, according to some embodiments, the immersion cooling system **1** (shown in FIG. **1**) that can prevent from the problems encountered by the cooling device known to the inventor (which is outside the system) is provided.

(30) For example, in FIG. **7**, the first communication pipeline **12** comprises a first main pipeline **127** and a plurality of first sub-pipelines **128**, **128'**. A first end of each of the first sub-pipelines **128**, **128'** is in communication with the first main pipeline **127**, so that the first sub-pipelines **128**, **128'** may be in communication with the first heat exchange device **13**. The second end of the first sub-pipeline **128** is in communication with the outlet **201**, so that the first sub-pipeline **128** may be in communication with the first microchannel **110** through the outlet **201**. The second end of the first sub-pipeline **128'** is in communication with the outlet **201'**, so that the first sub-pipeline **128'** may be in communication with the first microchannel **110'** through the outlet **201'**. The diameters of the first main pipeline **127** and the first sub-pipelines **128**, **128'** are not limited; for example, the diameters of the first sub-pipelines **128**, **128'** are both less than the diameter of the first main pipeline **127**. The materials of the first main pipeline **127** and the first sub-pipelines **128**, **128'** may independently comprise at least one selected from the group consisting of metal, alloy, ceramic, plastic, and rubber, and the materials of the first main pipeline **127** and the first sub-pipelines **128**, **128'** may be identical to or different from each other, both of which are not limited herein.

Moreover, in this embodiment, the immersion cooling system **1** may comprise the second communication pipeline **14**, or comprise the second communication pipeline **14** and the pump **15**, or comprise the second communication pipeline **14**, the pump **15**, and the third communication pipeline **16**, detail configurations of the implementations can be referred to above descriptions and thus will not be further described in detail herein. Therefore, through various configurations of the inlets **200**, **200'** and the outlets **201**, **201'**, according to some embodiments, the immersion cooling system **1** (shown in FIG. **1**) that can prevent from the problems encountered by the cooling device known to the inventor (which is outside the system) is provided.

(31) For another example, in FIG. **7**, the immersion cooling system **1** further comprises a second

communication pipeline **14**, wherein the second communication pipeline **14** is in the fluid section **L** and the second communication pipeline **14** comprises a second main pipeline **140** and a plurality of second sub-pipelines **141**, **141'**. The second sub-pipeline **141** is in communication with the inlet **200**, so that the second sub-pipeline **141** may be in communication with the second microchannel **111** through the inlet **200**. The second sub-pipeline **141'** is in communication with the inlet **200'**, so that the second sub-pipeline **141'** may be in communication with the second microchannel **111'** through the inlet **200'**. The diameters of the second main pipeline **140** and the second sub-pipelines **141**, **141'** are not limited; for example, the diameters of the second sub-pipelines **141**, **141'** are both less than the diameter of the second main pipeline **140**. The materials of the second main pipeline **140** and the second sub-pipelines **141**, **141'** may independently comprise at least one selected from the group consisting of metal, alloy, ceramic, plastic, and rubber, and the materials of the second main pipeline **140** and the second sub-pipelines **141**, **141'** may be identical to or different from each other, both of which are not limited herein. Moreover, in this embodiment, the immersion cooling system **1** may comprise the pump **15** or comprise the pump **15** and the third communication pipeline **16**, detail configurations of the implementations can be referred to above descriptions and thus will not be further described in detail herein. Therefore, through various configurations of the inlets **200**, **200'** and the outlets **201**, **201'**, according to some embodiments, the immersion cooling system **1** (shown in FIG. 1) that can prevent from the problems encountered by the cooling device known to the inventor (which is outside the system) is provided.

(32) Please refer to both FIG. 1 and FIG. 5. FIG. 5 illustrates a schematic perspective view of a partial section A of the immersion cooling system **1** shown in FIG. 1 according to a fifth embodiment. The heat exchange channels **131** of the first heat exchange device **13** in general have a flow direction **D2** (shown in FIG. 1; or the flow direction **D2'** shown in FIG. 5). According to various demands, the flow directions **D2**, **D2'** may be configured to have different angles with respect to the vertical line (e.g., the Z directions shown in FIG. 1 and FIG. 5). For example, in FIG. 1, the first heat exchange device **13** is adjacent to the chip device **2**, and the flow direction **D2** of the heat exchange channel **131** is substantially parallel to the vertical line (that is, an angle is between the flow direction **D2** and the vertical line, and the angle is less than, for example, 45 degrees). Therefore, after the circulation fluid **120** (or the liquid-phase working fluid **101**) is discharged from the outlet **201**, the circulation fluid **120** (or the liquid-phase working fluid **101**) can directly enter the interior of the heat exchange channels **131** to conduct heat exchange rapidly. Alternatively, for another example, in FIG. 5, the length of the first communication pipeline **12** is relatively longer, and thus the first heat exchange device **13** is configured away from the chip device **2**, and the flow direction **D2'** of the heat exchange channel **131** is substantially perpendicular to the vertical line (that is, an angle is between the flow direction **D2'** and the vertical line, and the angle is, for example, 90 degrees±45 degrees). Therefore, after the circulation fluid (or the liquid-phase working fluid **101**) is discharged from the outlet **201**, the circulation fluid **120** (or the liquid-phase working fluid **101**) can conduct heat exchange thoroughly in the relatively longer first communication pipeline **12**, and thus the circulation fluid **120** (or the liquid-phase working fluid **101**) with a lower temperature (i.e., a greater temperature difference is between the circulation fluid **120** and the first heat exchange device **13**) may enter the first heat exchange device **13**, thereby reducing the work loading of the first heat exchange device **13**.

(33) Please refer to FIG. 1 and FIG. 8A to FIG. 8C. FIG. 8A to FIG. 8C illustrate cross-sectional schematic views of the first communication pipeline **12** according to some embodiments. In FIG. 1, the interior of the first communication pipeline **12** has an inner wall **122** (shown in FIG. 8A to FIG. 8C). The first communication pipeline **12** comprises a pipeline disturbing portion **123**, and the pipeline disturbing portion **123** is on the inner wall **122** (shown in FIG. 8A to FIG. 8C). The pipeline disturbing portion **123** may be a mesh **124** (shown in FIG. 8A), metal particles **125** (shown in FIG. 8B), a fin **126** (shown in FIG. 8C), or a combination thereof, which will be described later. In other words, the pipeline disturbing portion **123** may comprise two or three of the mesh **124**, the

metal particles **125**, and the fin **126**; for example, the pipeline disturbing portion **123** may comprise a combination of the mesh **124** and the metal particles **125**, or a combination of the mesh **124** and the fin **126**, or a combination of the metal particles **125** and the fin **126**, or a combination of the mesh **124**, the metal particles **125**, and the fin **126**. Through the pipeline disturbing portion **123** on the inner wall **122** of the first communication pipeline **12**, when the circulation fluid **120** flows by the first communication pipeline **12**, bubbles **121** (i.e., the circulation fluid **120** in gas phase) will be formed on the interface between the circulation fluid **120** (which has a liquid-phase surface) and the pipeline disturbing portion **123** (which has, for example, a solid-phase surface) for the heterogeneous nucleation on the interface. The more the bubbles **121** are formed, the more the heats are removed from the liquid-phase circulation fluid **120**, and thus the liquid-phase circulation fluid **120** can be cooled more properly.

(34) For example, in FIG. **8A**, the pipeline disturbing portion **123** comprises the mesh **124**. The mesh **124** may be in one or more layers, which is not limited; for example, in FIG. **8A**, the mesh **124** is in four to five layers. The holes of the mesh **124** and the number of the holes per unit area are not limited and may be configured according to various demands. The materials of the mesh **124** may be metal, alloy, or a combination thereof, which is not limited herein. Accordingly, through the meshes **124** on the inner wall **122** of the first communication pipeline **12**, a rougher surface with more cavities and gaps can be formed on the inner wall **122**. Therefore, the heterogeneous nucleation can be formed on the rougher surface with more cavities and gaps by the circulation fluid **120**, which may be beneficial to the heterogeneous nucleation, and thus forming more bubbles **121**.

(35) For another example, in FIG. **8B**, the pipeline disturbing portion **123** comprises the metal particles **125**. The size and the size distribution of the metal particles **125** are not limited and may be configured according to various demands. The size of the metal particles **125** may be substantially identical to or different from each other, which is not limited herein. The materials of the metal particles **125** may be metal, alloy, or a combination thereof, which is not limited herein; for example, the material of the metal particles **125** is copper. Accordingly, through the metal particles **125** on the inner wall **122** of the first communication pipeline **12**, a rougher surface with more cavities and gaps may be formed on the inner wall **122**. Therefore, the heterogeneous nucleation can be formed on the rougher surface with more cavities and gaps by the circulation fluid **120**, which may be beneficial to the heterogeneous nucleation, and thus forming more bubbles **121**. Moreover, in some embodiments, the stacking height of the metal particles **125** (e.g., the height along the Y direction shown in FIG. **8B**) may be configured according to various demands; for example, in FIG. **8B**, the metal particles **125** on the side of the inner wall **122** adjacent to the outlet **201** (shown in FIG. **1**) (that is, the side adjacent to the -Z direction shown in FIG. **8B**) are stacked in a greater stacking height, so that more heterogeneous nucleation may be formed by the circulation fluid **120** once the circulation fluid **120** enters the first communication pipeline **12**. Hence, the bubbles **121** can be formed more rapidly by the circulation fluid **120**, and thus the circulation fluid **120** from the chip device **2** (shown in FIG. **1**) can be cooled more properly.

(36) For another example, in FIG. **8C**, the pipeline disturbing portion **123** comprises the fin **126**. The fin **126** may be one or more, which is not limited; for example, the fin **126** may be a fin array (shown in FIG. **8C**). The extending direction of the fin **126** is not limited. For example, in FIG. **8C**, a plurality of the fins **126** are aligned along an alignment direction (e.g., the +Z direction or the flow direction D2 shown in FIG. **8C**), and the extending direction of each of the fins **126** (e.g., the +Y direction shown in FIG. **8C**) is substantially perpendicular to the flow direction D2 of the heat exchange channel **131** (that is, in this embodiment, the flow direction D2 of the circulation fluid **120** inside the heat exchange channel **131**). Alternatively, for another example, a plurality of the fins **126** are aligned along an alignment direction (e.g., the circumferential direction of the inner wall **122** of the first communication pipeline **12**), and the extending direction of each of the fins **126** (e.g., the +Z direction shown in FIG. **8C**) is substantially parallel to the flow direction of the

circulation fluid **120** (e.g., the +Z direction or the flow direction D2 shown in FIG. 8C). In other words, the length (e.g., along the Z direction shown in FIG. 8C), the width (e.g., along the X direction shown in FIG. 8C), and the height (e.g., along the Y direction shown in FIG. 8C) of each of the fins **126** may be configured according to various demands, and may respectively be substantially identical to or different from each other, both of which are not limited herein. The pitch between any two of the fins **126** is not limited herein. The materials of the fins **126** may independently be metal, alloy, or a combination thereof, which is not limited herein. Accordingly, through the fin(s) **126** on the inner wall **122** of the first communication pipeline **12**, a rougher surface with more cavities and gaps can be formed on the inner wall **122**. Therefore, the heterogeneous nucleation can be on the rougher surface with more cavities and gaps by the circulation fluid **120**, which may be beneficial to the heterogeneous nucleation, and thus forming more bubbles **121**. Moreover, in some embodiments, the height of the fin(s) **126** may be configured according to various demands; for example, in FIG. 8C, the fins **126** on the side of the inner wall **122** adjacent to the outlet **201** (shown in FIG. 1) (that is, the side adjacent to the -Z direction shown in FIG. 8C) are in a greater height, so that more heterogeneous nucleation may be formed by the circulation fluid **120** once the circulation fluid **120** enters the first communication pipeline **12**. Hence, the bubbles **121** can be formed more rapidly by the circulation fluid **120**, and thus the circulation fluid **120** from the chip device **2** (shown in FIG. 1) can be cooled more properly.

(37) Referring back to FIG. 1, in some embodiments, the immersion cooling system **1** further comprises a cooling device **17**, wherein the cooling device **17** is at the side of the chip device **2** away from the motherboard **20**; for example, the cooling device **17** is at the side of the cover **22** away from the motherboard **20**. The cooling device **17** may be any device that can conduct heat exchange, such as a boiler plate. The materials of the cooling device **17** may be any material with thermal conductive function, which is not limited herein; for example, the materials may be metal, alloy, or a combination thereof. Therefore, through the cooling device **17** that is substantially contacted with the chip device **2** and/or the chip(s) **21** of the chip device **2**, the cooling device **17** may conduct heat exchange more properly with the chip device **2** and/or the chip(s) **21** of the chip device **2**. Moreover, in some embodiments, one side of the cooling device **17** (e.g., the side adjacent to the motherboard **20**) is substantially contacted with the chip device **2** and/or the chip(s) **21** of the chip device **2**, and another side of the cooling device **17** (e.g., the side away from the motherboard **20**) is substantially contacted with the liquid-phase working fluid **101**. Accordingly, through the cooling device **17**, the heat of the chip device **2** and/or the chip(s) **21** of the chip device **2** can be removed and thermally conducted more directly and rapidly with the liquid-phase working fluid **101**, so that the chip device **2** and/or the chip(s) **21** of the chip device **2** can be cooled more efficiently.

(38) For example, referring to FIG. 1 and FIG. 9A to FIG. 9C, FIG. 9A to FIG. 9C illustrate cross-sectional schematic views of the cooling device **17** according to some embodiments. In FIG. 1, the cooling device **17** has a cooling surface **170** (shown in FIG. 9A to FIG. 9C), and the cooling surface **170** is at the side of the cooling device **17** away from the motherboard **20**. The cooling device **17** comprises a cooling disturbing portion **171**, and the cooling disturbing portion **171** is on the cooling surface **170** (shown in FIG. 9A to FIG. 9C). The cooling disturbing portion **171** may be a mesh **172** (shown in FIG. 9A), metal particles **173** (shown in FIG. 9B), a fin **174** (shown in FIG. 9C), or a combination thereof, which will be described later. In other words, the cooling disturbing portion **171** may comprise two or three of the mesh **172**, the metal particles **173**, and the fin **174**; for example, the cooling disturbing portion **171** may comprise a combination of the mesh **172** and the metal particles **173**, or a combination of the mesh **172** and the fin **174**, or a combination of the metal particles **173** and the fin **174**, or a combination of the mesh **172**, the metal particles **173**, and the fin **174**. Through the cooling disturbing portion **171** on the cooling surface **170** of the cooling device **17**, when the liquid-phase working fluid **101** flows to the proximity of the cooling device **17**, bubbles **103** (i.e., the gas-phase working fluid) will be formed on the interface between the

liquid-phase working fluid **101** (which has a liquid-phase surface) and the cooling disturbing portion **171** (which has, for example, a solid-phase surface) for the heterogeneous nucleation on the interface, and thus the liquid-phase working fluid **101** can be cooled more properly.

(39) For example, in FIG. **9A**, the cooling disturbing portion **171** comprises the mesh **172**. The mesh **172** may be in one or more layers, which is not limited; for example, in FIG. **9A**, the mesh **172** is in four to five layers. The holes of the mesh **172** and the number of the holes per unit area are not limited and may be configured according to various demands. The materials of the mesh **172** may be metal, alloy, or a combination thereof, which is not limited herein. Accordingly, through the meshes **172** on the cooling surface **170** of the cooling device **17**, a rougher surface with more cavities and gaps can be formed on the inner wall **122**. Therefore, the heterogeneous nucleation can be formed on the rougher surface with more cavities and gaps by the liquid-phase working fluid **101**, which may be beneficial to the heterogeneous nucleation, and thus forming more bubbles **103**.

(40) For another example, in FIG. **9B**, the cooling disturbing portion **171** comprises the metal particles **173**. The size and the size distribution of the metal particles **173** are not limited and may be configured according to various demands. The size of the metal particles **173** may be substantially identical to or different from each other, which is not limited herein. The materials of the metal particles **173** may be metal, alloy, or a combination thereof, which is not limited herein; for example, the material of the metal particles **173** is copper. Accordingly, through the metal particles **173** on the cooling surface **170** of the cooling device **17**, a rougher surface with more cavities and gaps may be formed on the cooling surface **170**. Therefore, the heterogeneous nucleation can be formed on the rougher surface with more cavities and gaps by the liquid-phase working fluid **101**, which may be beneficial to the heterogeneous nucleation, and thus forming more bubbles **103**. Moreover, in some embodiments, the stacking height (e.g., the height along the Y direction shown in FIG. **9B**) of the metal particles **173** may be configured according to various demands; for example, in FIG. **9B**, the metal particles **173** on the side of the cooling surface **170** towards the gravity direction (e.g., the $-Z$ direction shown in FIG. **9B**) are stacked in a greater stacking height, so that more heterogeneous nucleation may be formed on the side towards the gravity direction (e.g., the side that is away from the flow direction of the bubbles **103**) by the liquid-phase working fluid **101**. Hence, the bubbles **103** can be formed more rapidly by the liquid-phase working fluid **101**. Furthermore, some of the bubbles **103** may be combined to each other to form larger bubbles when flowing to the surface of the liquid-phase working fluid **101**, so that much more heat can be removed from the liquid-phase working fluid **101**. Accordingly, according to some embodiments, the liquid-phase working fluid **101** in a wider range can be cooled by the cooling disturbing portion **171** at one time, thereby enhancing the cooling performance of the liquid-phase working fluid **101**.

(41) For another example, in FIG. **9C**, the cooling disturbing portion **171** comprises the fin **174**. The fin **174** may be one or more, which is not limited; for example, the fin **174** may be a fin array (shown in FIG. **9C**). The extending direction of the fin **174** is not limited. For example, in FIG. **9C**, a plurality of the fins **174** are aligned along an alignment direction (e.g., the $+Z$ direction shown in FIG. **9C**), and the extending direction (e.g., the $+Y$ direction shown in FIG. **9C**) of each of the fins **174** is substantially perpendicular to the flow direction of the liquid-phase working fluid **101** (e.g., the $+Z$ direction shown in FIG. **9C**). Alternatively, for another example, a plurality of the fins **174** are aligned along an alignment direction (e.g., the X direction shown in FIG. **9C**), and the extending direction of each of the fins **174** (e.g., the $+Z$ direction shown in FIG. **9C**) is substantially parallel to the flow direction of the liquid-phase working fluid **101** (e.g., the $+Z$ direction shown in FIG. **9C**). In other words, the length (e.g., along the Z direction shown in FIG. **9C**), the width (e.g., along the X direction shown in FIG. **9C**), and the height (e.g., along the Y direction shown in FIG. **9C**) of each of the fins **174** may be configured according to various demands, and may respectively be substantially identical to or different from each other, which is not limited herein. The pitch

between any two of the fins **174** is not limited herein. The materials of the fins **174** may independently be metal, alloy, or a combination thereof, which is not limited herein. Accordingly, through the fin(s) **174** on the cooling surface **170** of the cooling device **17**, a rougher surface with more cavities and gaps can be formed on the cooling surface **170**. Therefore, the heterogeneous nucleation can be formed on the rougher surface with more cavities and gaps by the liquid-phase working fluid **101**, which is beneficial to the heterogeneous nucleation, and thus forming more bubbles **103**. Moreover, in some embodiments, the height of the fin(s) **174** may be configured according to various demands; for example, in FIG. **9C**, the fins **174** on the side of the cooling surface **170** towards the gravity direction (e.g., the $-Z$ direction shown in FIG. **9C**) are in a greater height, so that more heterogeneous nucleation may be formed on the side towards the gravity direction (e.g., the side that is away from the flow direction of the bubbles **103**) by the liquid-phase working fluid **101**. Hence, the bubbles **103** can be formed more rapidly by the liquid-phase working fluid **101**. Furthermore, some of the bubbles **103** may be combined to each other to form larger bubbles when flowing to the surface of the liquid-phase working fluid **101**, so that much more heat can be removed from the liquid-phase working fluid **101**. Accordingly, according to some embodiments, the liquid-phase working fluid **101** in a wider range can be cooled by the cooling disturbing portion **171** at one time, thereby enhancing the cooling performance of the liquid-phase working fluid **101**.

(42) To sum up, in some embodiments, by configuring a chip device to be in communication with cooling pipelines (e.g., a microchannel device, a first communication pipeline, and a first heat exchange device) that are also inside the system (or inside the work tank of the system), according to some embodiments, the cooling pipelines of an immersion cooling system can be prevented from being in communication with the exterior of the system (or the exterior of the work tank of the system). Therefore, according to some embodiments, through the cooling pipelines inside the system (and thus the length of the cooling pipeline may be much shorter as compared to that of the prior arts known to the inventor), the immersion cooling system can still achieve the cooling performance substantially equal to or even superior to that of the prior arts known to the inventor by consuming a less pumping power.

Claims

1. An immersion cooling system comprising: a work tank comprising a fluid section; a chip device in the fluid section and having an inlet and an outlet, wherein the chip device comprises: a motherboard having a main surface substantially parallel to a vertical line; a chip; and a cover on the motherboard, wherein the chip is between the cover and the motherboard; a microchannel device in the chip device, wherein a first end of the microchannel device is in communication with the inlet, and wherein a second end of the microchannel device is in communication with the outlet; a first communication pipeline in the fluid section, wherein a first end of the first communication pipeline is in communication with the outlet; and a first heat exchange device in the fluid section and in communication with a second end of the first communication pipeline-; wherein the work tank further comprises a vapor section, and the immersion cooling system further comprises a second heat exchange device in the vapor section.
2. The immersion cooling system according to claim 1, wherein the inlet is selectively on the motherboard, or on the cover, or on both the motherboard and the cover; and the outlet is selectively on the motherboard, or on the cover, or on both the motherboard and the cover.
3. The immersion cooling system according to claim 2, wherein: the inlet is on the motherboard or the cover; the outlet is on the motherboard or the cover; and the microchannel device comprises: a first microchannel, wherein a first end of the first microchannel is in communication with the chip, a second end of the first microchannel is in communication with the outlet, and a diameter of the first microchannel is less than a diameter of the first communication pipeline; and a second

- microchannel, wherein a first end of the second microchannel is in communication with the chip, and a second end of the second microchannel is in communication with the inlet.
4. The immersion cooling system according to claim 3, further comprising a second communication pipeline, wherein the second communication pipeline is in the fluid section and in communication with the inlet.
 5. The immersion cooling system according to claim 4, further comprising a pump, wherein the pump is in the fluid section and in communication with the second communication pipeline.
 6. The immersion cooling system according to claim 5, wherein the chip device is electrically connected to the pump to supply an electricity for the pump and control the pump.
 7. The immersion cooling system according to claim 6, wherein the pump is on the motherboard to be electrically connected to the motherboard.
 8. The immersion cooling system according to claim 5, further comprising a third communication pipeline, wherein the third communication pipeline is in the fluid section, a first end of the third communication pipeline is in communication with the first heat exchange device, and a second end of the third communication pipeline is in communication with the pump.
 9. The immersion cooling system according to claim 3, further comprising a cooling device, wherein the cooling device is at a side of the chip device away from the motherboard.
 10. The immersion cooling system according to claim 9, wherein the cooling device is at a side of the cover away from the motherboard.
 11. The immersion cooling system according to claim 10, wherein the cooling device has a cooling surface, the cooling surface is at a side of the cooling device away from the motherboard, and the cooling device comprises a cooling disturbing portion on the cooling surface.
 12. The immersion cooling system according to claim 11, wherein the cooling disturbing portion is a mesh, a plurality of metal particles, a fin, or a combination comprising two or more of the mesh, the metal particles, and the fin.
 13. The immersion cooling system according to claim 3, wherein the first communication pipeline has an inner wall, and the first communication pipeline comprises a pipeline disturbing portion on the inner wall.
 14. The immersion cooling system according to claim 13, wherein the pipeline disturbing portion is a mesh, a plurality of metal particles, a fin, or a combination comprising two or more of the mesh, the metal particles, and the fin.
 15. The immersion cooling system according to claim 3, wherein the first heat exchange device comprises a plurality of heat exchange channels, and a flow direction of each of the heat exchange channels is substantially perpendicular to the vertical line.
 16. The immersion cooling system according to claim 3, wherein the first heat exchange device comprises a plurality of heat exchange channels, and a flow direction of each of the heat exchange channels is substantially parallel to the vertical line.
 17. The immersion cooling system according to claim 3, wherein the work tank further comprises a vapor section, the immersion cooling system further comprises a second heat exchange device, and the second heat exchange device comprises: a condenser in the vapor section; a condensing pump; and a heat exchanger, wherein a first end of the condensing pump is in communication with a first end of the condenser, a second end of the condensing pump is in communication with a first end of the heat exchanger, and a second end of the heat exchanger is in communication with a second end of the condenser.
 18. The immersion cooling system according to claim 17, wherein the condensing pump is outside the work tank.
 19. The immersion cooling system according to claim 17, wherein the heat exchanger is outside the work tank.
 20. The immersion cooling system according to claim 3, wherein the first heat exchange device comprises a partition plate and a plurality of heat exchange channels, and the heat exchange

channels are partitioned by the partition plate.

21. The immersion cooling system according to claim 2, wherein: the inlet is on both the motherboard and the cover; the outlet is on both the motherboard and the cover; and the microchannel device comprises: a plurality of first microchannels, wherein a first end of each of the first microchannels is in communication with the chip, a second end of each of the first microchannels is in communication with the outlet, and a diameter of each of the first microchannels is less than a diameter of the first communication pipeline; and a plurality of second microchannels, wherein a first end of each of the second microchannels is in communication with the chip, and a second end of each of the second microchannels is in communication with the inlet.
